ACTL3182 Wk5

Tadhg XG

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Forwards

Forward Contract

A forward contract is an agreement between two parties where one will buy the underlying off the other at an agreed upon price F_0 at time T.

Under no arbitrage, the forward price must be

$$F_0 = \text{Current spot price} + \text{Cost of carry} - \text{Gains of carry}.$$

E.g. If a non-dividend paying stock is worth S_0 now, then a 1-year forward price is

$$F_0 = S_0 + i \cdot S_0 = S_0(1 + i).$$

Options

Calls and Puts

A European option gives the buyer the right, but not the obligation, to buy (a call) or sell (a put) the underlying at time T at a price K.

The payoff of a call is thus

$$(S_T - K)^+ = \max(S_T - K, 0) = (S_T - K)\mathbb{1}(S_T > K),$$

and the payoff a put is

$$(K - S_T)^+ = \max(K - S_T, 0) = (K - S_T)\mathbb{1}(K > S_T).$$

Put-Call Parity

Put-Call Parity Theorem

Let c_t, p_t be the price of a call and put at time t respectively on a stock worth S_t, expiring at time T. Then,

$$c_t + Ke^{-r(T-t)} = p_t + S_t.$$

Proof of PC Parity

Proof: Consider two portfolios,

A: One call and $Ke^{-r(T-t)}$ cash (to finance the exercise);

B: One put and the stock (so we can actually exercise it).

Then, at time T the value of each portfolio is

$$A = \begin{cases} S_T - K + K = S_T, & \text{if } S_T > K, \\ 0 + K = K, & \text{if } S_T < K, \end{cases}$$

and

$$B = \begin{cases} 0 + S_T = S_T, & \text{if } S_T > K, \\ K - S_T + S_T, & \text{if } S_T < K. \end{cases}$$

Since the portfolios have the same end value regardless of future price, by no arbitrage they must be worth the same. Hence,

$$c_t + Ke^{-r(T-t)} = p_t + S_t.$$

Probability revisited

Let Ω be a sample space and $\mathcal{F}$ a collection of events (a σ-algebra).

A probability measure $\mathbb{P}$ needs to satisfy

$$\mathbb{P}(A) \in [0, 1], \quad \forall A \in \mathcal{F},$$
$$\mathbb{P}(\Omega) = 1.$$

Intuitively, the probabilities assigned just need to be between 0 and 1 and the probability of anything happening needs to be 1. Hence, there are multiple possible measures for a single sample space, say $\mathbb{P}$ and $\mathbb{Q}$.

Binomial Model

Under a discrete time setting, suppose at time t with a stock currently at S_t, which behaves under $\mathbb{P}$ as

$$S_{t+1} = \begin{cases} uS_t, & \text{w.p. } p, \\ dS_t, & \text{w.p. } 1-p. \end{cases}$$

Then, the value of a call maturing in 1 period of length δ is

$$c_t = e^{-r\delta}\,\mathbb{E}_{\mathbb{Q}}[(S_T - K)^+] = e^{-r\delta}\left(q(uS_t - K)^+ + (1-q)(dS_t - K)^+\right),$$

where $\mathbb{Q}$ is the risk-neutral measure which assigns

$$\mathbb{Q}(\text{stock goes up}) = q = \frac{e^{r\delta} - d}{u - d}.$$

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This can be extended to multiple periods and any derivative payoff.

Proof of $\mathbb{Q}$

More generally, say the derivative has payoff X and will be either c_u or c_d in one period, and we can buy/sell as much of the stock and invest risk free.

We want to buy ϕ stock and ψ bond such that it replicates the portfolio:

$$\begin{cases} \phi u S_0 + \psi e^{r\delta} = c_u, \\ \phi d S_0 + \psi e^{r\delta} = c_d. \end{cases}$$

Rearranging and solving for ϕ and ψ, we get

$$\phi = \frac{c_u - c_d}{S_0(u-d)}, \quad \psi = e^{-r\delta}\left(\frac{c_d u - c_u d}{u-d}\right)$$

Proof of $\mathbb{Q}$ (cont.)

So, the cost of this portfolio is

$$\begin{aligned}
\phi S_0 + \psi &= \frac{c_u - c_d}{S_0(u-d)} S_0 + e^{-r\delta}\left(\frac{c_d u - c_u d}{u-d}\right) \\
&= e^{-r\delta}\left(\frac{e^{r\delta}c_u - e^{r\delta}c_d}{u-d} + \frac{c_d u - c_u d}{u-d}\right) \\
&= e^{-r\delta}\left\{c_u \underbrace{\frac{e^{r\delta}-d}{u-d}}_{=q} + c_d \underbrace{\frac{u-e^{r\delta}}{u-d}}_{=1-q}\right\} \\
&= e^{-r\delta}\,\mathbb{E}_{\mathbb{Q}}[X],
\end{aligned}$$

which by no arbitrage is the price of the derivative.

Remark

By no arbitrage, we must have

$$q = \frac{e^{r\delta} - d}{u - d} \in [0, 1],$$

and so technically these q values are probabilities (though not *likelihoods*) for the stock going up or down.

Martingale

Recall a stochastic process is a series of random variables indexed by t.

Definition (Martingale)

Let $\mathcal{F}(t)$ be the (random) history of stock prices up to time t. A stochastic process $M(\cdot)$ is said to be a $\mathbb{Q}$-martingale if

$$\mathbb{E}_{\mathbb{Q}}[M(u) \mid \mathcal{F}(t)] = M(t), \quad \forall u \geq t.$$

That is, given all the information up to time t, our best guess of wherever M will be in the future is just what it is right now.

Crucially, both sides are random variables!!